Probability: Randomness, Conditioning, and Bayes

Unit 7: What Chance Produces, and How Evidence Moves You

Stat 220 | Statistics for Data Science

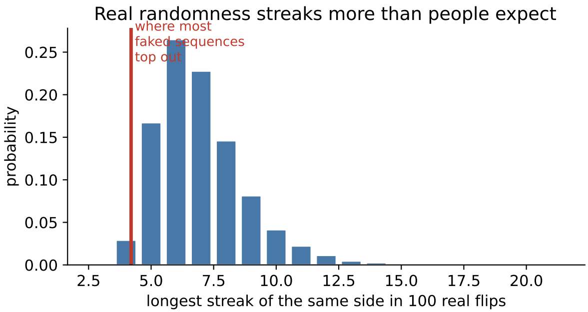
The thing we assumed last unit

- Unit 1 kept asking: *what would this look like if nothing were going on?* We answered it by hand for the Coke test and then waved at it everywhere else.
- To answer it in general, we need the machinery: how to describe a random process, combine events, and attach a number to an uncertain outcome.
- That is this unit. Three days: what randomness looks like, how evidence updates it, and how to summarize an uncertain quantity with a mean and a spread.

Principle: probability is a claim about a process, not about one outcome

"There is a 30% chance of rain" is a statement about how often days like this one end in rain. No single afternoon can prove it right or wrong.

The giveaway is the streaks



In my 120 pre-flipped sequences the longest streak averaged **5.8**, and ran as high as **12**.

- Invented sequences top out around 4.
- Real randomness **clumps**. Invented sequences spread out too evenly, and that is what gives them away.

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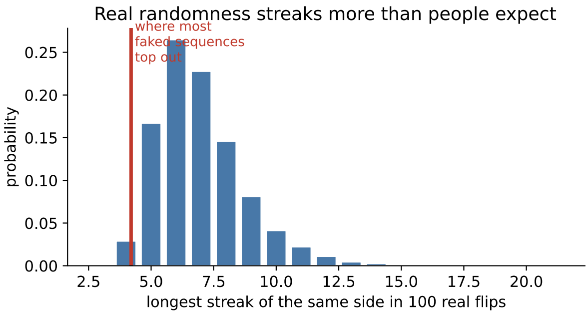

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Count how many I got right. This is not a superpower: I know the distribution of a streak length, and you were guessing.

Why that game matters at work

- Three support tickets in one week from the same region. A pattern, or a clump?
- Four straight bad quarters for one sales rep. A slump, or a streak?
- A cluster of server failures on Tuesdays. A cause, or a coincidence?

Principle: the null hypothesis of everyday life

Before you explain a pattern, work out what pure chance would have produced. Most "patterns" in small data are exactly the clumping you failed to imagine.

What this unit gives you

Things to know

- the rules for combining events, and what independence really requires
- conditional probability, and why $P(A \mid B) \neq P(B \mid A)$
- Bayes' rule, and the decisive role of the base rate
- expectation and variance, and what happens when you add things up
- which standard distribution matches which real-world story

Mistakes to stop making

- flipping a conditional probability around
- ignoring the base rate after a positive test
- multiplying probabilities that were never independent
- expecting chance to "even out" in the short run
- reporting a mean with no spread, or calling it typical when the data is skewed
- plugging averages into a nonlinear calculation

The vocabulary, quickly

- An **experiment** is any process with an uncertain outcome: a flip, a customer visit, a quarter of sales.
- The **sample space** is everything that could happen.
- An **event** is a subset of that: "the customer buys," "revenue tops $1M."
- $P(A)$ is a number in $[0, 1]$ measuring how often A happens if the process runs many times.

Principle: name the sample space first

Most probability errors are not arithmetic errors. They are errors about *which set of outcomes we are talking about.*

Three rules that generate everything else

1. **Complement:** $P(\text{not } A) = 1 - P(A)$.
2. **Addition:** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. Subtract the overlap so it is not counted twice.
3. **Multiplication:** $P(A \text{ and } B) = P(A)\, P(B \mid A)$. *Always* true. That $P(B \mid A)$ is a **conditional** probability, defined on the next slide, and it collapses to $P(A)P(B)$ only when the two are independent.

Trap: the shortcut everyone takes

Multiplying $P(A)P(B)$ is the most common probability error in industry, because independence gets assumed silently instead of argued for.

The complement rule is your best friend

"At least one" problems are almost always easier backwards:

$$P(\text{at least one failure}) = 1 - P(\text{no failures}).$$

- A service fails on a given day with probability 0.01. Over a 90-day quarter, if days were independent, $P(\text{at least one outage}) = 1 - 0.99^{90} \approx \mathbf{0.60}$.
- A 1% daily risk is a coin flip over a quarter. Rare events stop being rare once you get many chances at them.

Principle: count the chances, not the impressions

"That basically never happens" is a statement about one trial. Always ask how many trials there are.

Conditional probability: shrinking the world

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

- Read it as: *throw away every outcome where B did not happen*, then ask how often A happens among what is left.
- Conditioning is what data science actually does. Every model output is a conditional statement: given these features, what do we expect?

Principle: a probability without a condition is incomplete

"What is the churn rate?" is a worse question than "what is the churn rate *for accounts under $500 a month in their first year?*"

Independence, stated carefully

A and B are **independent** when knowing one tells you nothing about the other:

$$P(A \mid B) = P(A).$$

- Two coin flips: independent.
- Two mortgages in the same city defaulting: not independent. They share a housing market.
- Two servers in the same rack failing: not independent. They share power, cooling, and a deploy pipeline.

Principle: ask what they share

Things are dependent when they share a cause. Go looking for the shared cause before you assume independence.

Why false independence is expensive

- In 2008, models priced pools of mortgages by treating individual defaults as nearly independent. Under that assumption, the chance that a huge fraction defaulted at once was effectively zero.
- Defaults were driven by a common factor, the national housing market. When it turned, they turned together.
- The model was not wrong about any single mortgage. It was wrong about the **dependence**, and that was enough to matter.

Trap: independence is an assumption, not a fact

Multiplying small probabilities makes catastrophes look impossible. That is exactly when to double-check whether the events share a driver.

Your turn #1

Your turn: is the multiplication legitimate?

An engineer says: "Each of our three data centers has a 1% chance of going down in a given month, so the chance all three are down at once is $0.01^3 =$ one in a million. We are fine."

Turn to a neighbor: what would have to be true for that number to be right, and would you bet the company on it?

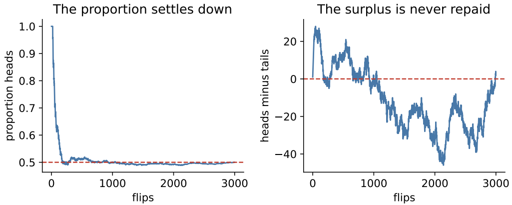
Your turn #1: answer

- The calculation requires the three outages to be **independent**.
- They are not. A bad deploy, an expired certificate, a cloud-provider outage, a BGP misconfiguration, or an attack hits all three at once.
- The honest statement: one in a million is a *lower bound under the most optimistic possible assumption*, and the real risk is dominated by the shared causes.

Principle: correlated failures are the ones that kill you

When you build redundancy you are betting on independence. Spend your effort making the failures independent, not on adding a fourth copy of the same thing behind the same load balancer.

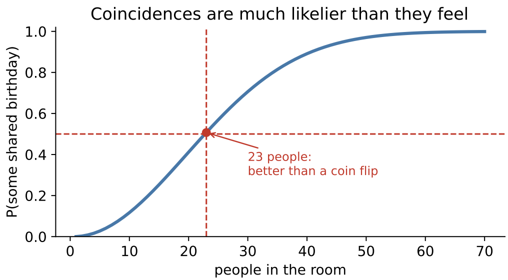
Two more things randomness does that surprise people



The coin is never "due." After a run of heads, the *proportion* drifts back toward one half because later flips swamp the early surplus. The surplus itself is never repaid, only diluted.

Small groups produce the extremes. The counties with the lowest cancer rates are mostly small rural ones. So are the counties with the highest. Same cause: tiny denominators.

Coincidences are more common than they feel



With 23 people a shared birthday is more likely than not. With 50 it is nearly certain.

- We intuit "someone shares *my* birthday," which is rare.
- The actual question involves all $\binom{23}{2} = 253$ *pairs*.

Same logic on your data: test 100 variables at the 0.05 level and about five look "significant" by construction.

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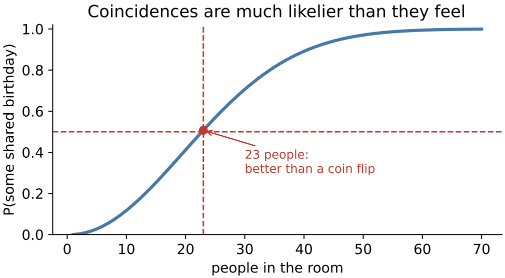

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Let us check: does anyone here share a birthday? Months first, then days.

The direction matters enormously

$P(A \mid B)$ and $P(B \mid A)$ are different numbers, often wildly different.

- $P(\text{speaks Spanish} \mid \text{lives in Spain})$ is near 1.
- $P(\text{lives in Spain} \mid \text{speaks Spanish})$ is small: most Spanish speakers live elsewhere.
- $P(\text{rich} \mid \text{owns a private jet})$ is near 1. The reverse, $P(\text{owns a private jet} \mid \text{rich})$, is not.

Trap: the prosecutor's fallacy

"The chance of this DNA match if he were innocent is 1 in a million, so there is a 1 in a million chance he is innocent." Those are different conditionals, and the second also depends on how many people could have been the source.

A wrongful conviction built on a flipped conditional

Setup. Sally Clark was convicted in the UK in 1999 after two of her infants died suddenly. An expert testified that the chance of two cot deaths in one family was "1 in 73 million," computed by squaring a single-death rate.

- **Error 1:** the two deaths were treated as independent, ignoring shared genetics and environment.
- **Error 2:** even taken at face value, that number is $P(\text{evidence} \mid \text{innocent})$, not $P(\text{innocent} \mid \text{evidence})$. The alternative, a mother murdering two infants, is *also* extremely rare, and the comparison is what matters.

The conviction was overturned in 2003, and the Royal Statistical Society issued a public statement about the misuse of the statistic.

Your turn #2

Your turn: translate the claim

A security vendor advertises: "Our system flags fraud with 95% accuracy."

With a neighbor, write down: (i) which conditional probability is 95% here, (ii) which conditional probability the fraud investigator actually cares about, and (iii) what extra number you need to get from one to the other.

Your turn #2: answer

- (i) The vendor means roughly $P(\text{flagged} \mid \text{fraud})$, a detection rate.
- (ii) The investigator, holding a flagged case, needs $P(\text{fraud} \mid \text{flagged})$. That is what decides whether the review queue is worth staffing.
- (iii) You need the **base rate** of fraud, plus the false-positive rate on the honest majority.

Principle: the missing number is almost always the base rate

Accuracy claims describe the test. Decisions need the rate of the thing being tested for.

Bayes' rule

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

- It is the formula for **reversing** a conditional, which is the operation we need constantly.
- $P(A)$ is the **prior**, what you believed beforehand. $P(A \mid B)$ is the **posterior**, what you believe after seeing the evidence.
- Unit 10 turns this into a whole way of doing inference. Today it is a tool for getting the direction right.

Principle: evidence updates a starting point, it does not replace one

A positive test moves your belief. Where it moves *to* depends on where it started.

The four numbers a test comes with

Any yes/no test has two error rates, and they have names you will meet constantly.

Name	**Conditional probability**	**In words**
Sensitivity	$P(\text{test}^+ \mid \text{sick})$	of those who have it, the share the test catches
Specificity	$P(\text{test}^- \mid \text{healthy})$	of those who do not, the share it clears
Prevalence	$P(\text{sick})$	how common it is to begin with
Predictive value	$P(\text{sick} \mid \text{test}^+)$	**what the patient wants to know**

Principle: the first three are given, the fourth is the answer

Sensitivity and specificity describe the test itself. The predictive value changes with the prevalence, which is why one test means different things in different populations.

Do not use the formula. Use 10,000 people.

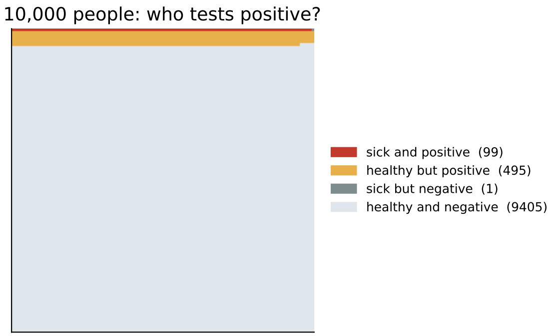
A disease affects **1%** of people (the prevalence). The test catches **99%** of sick people (sensitivity 0.99) and is wrong on **5%** of healthy people (specificity 0.95). You test positive. What is the chance you are sick?

Picture 10,000 people:

- 100 are sick. Of those, **99** test positive.
- 9,900 are healthy. Of those, **495** test positive anyway.
- So 594 people test positive, and only 99 of them are sick.

$$P(\text{sick} \mid \text{positive}) = \frac{99}{99 + 495} \approx \mathbf{17\%}.$$

The same fact, as a picture



The healthy group is so much larger that its 5% error rate produces **five times** as many positives as the disease itself.

A "99% accurate" test for a rare condition is mostly a false-positive machine. No amount of rhetoric changes the arithmetic.

Why counting people beats the formula

- Natural frequencies (99 out of 594) are far easier for humans, including doctors and executives, than conditional probabilities.
- In studies, physicians given percentages get this problem wrong most of the time. Given counts of patients, most get it right.
- Drawing the 10,000-person table is a better answer than reciting Bayes' rule, because it also *explains* the result to whoever asked.

Principle: the move to reach for

Whenever someone hands you a sensitivity, a specificity, and a rare condition, immediately build a table of a big round number of people.

This is the churn model, the fraud flag, the spam filter

- Your model flags 500 accounts as "about to churn." If churn is 2% and your false-positive rate is 5%, most of that list is not churning.
- That is not a broken model. It is what a rare outcome does to *any* imperfect classifier.
- The fix is often not a better model but a better **base rate**: run the classifier only on a segment where the outcome is common.

You may be asked: a standard screening question

"Your model is 99% accurate at predicting a 1-in-1000 event. Are you impressed?" No: predicting "never" scores 99.9%. Unit 8 builds the full vocabulary for this.

Your turn #3

Your turn: run the numbers yourself

A drug test is 98% accurate in both directions: it catches 98% of users and clears 98% of non-users. In a workplace where **0.5%** of employees actually use the drug, an employee tests positive.

Take 10,000 employees. How many test positive? What fraction of them are real users?

Your turn #3: answer

Out of 10,000 employees:

- 50 are users, and 98% of them, about **49**, test positive.
- 9,950 are not, and 2% of them, **199**, test positive anyway.
- Positives: $49 + 199 = 248$, of whom 49 are real users. About **20%**.

Trap: a defensible test, an indefensible decision

Firing everyone who tests positive means firing four innocent people for every real user. The test is fine. Using it alone as a decision rule is not.

LLM check: mistake or not? (1 of 2)

LLM check: we asked an LLM about a screening result

We told it: "A test is 99% accurate. The disease affects 1 in 1000 people. I tested positive. What is the chance I have it?"

It answered: "The test is 99% accurate, so there is about a 99% chance you have the disease. You should follow up with your doctor immediately."

Mistake, or no mistake?

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Mistake, or no mistake?

Trap: verdict: mistake

It ignored the base rate. In 100,000 people, 100 are sick (99 test positive) and 99,900 are healthy (about 999 test positive anyway). The real answer is near **9%**, not 99%.

LLM check: mistake or not? (2 of 2)

LLM check: a question about combining risks

We told it: "Our two suppliers each fail 3% of the time. What is the chance both fail in the same week?"

It answered: "If the failures were independent it would be about 0.09%, but I would not rely on that. If both ship through the same port or region, a storm or a strike takes out both at once, and the true joint probability could be many times higher. I would ask what the two suppliers have in common before quoting a number."

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Mistake, or no mistake?

Principle: verdict: no mistake

It gives the independent calculation, labels the assumption, and goes looking for the shared cause. That is the habit from Day 1, applied correctly.

Turning a probability question into arithmetic

1. **Name the events in words** and write which conditional you were given.
2. **Ask which one you were asked for.** If it is the other direction, you need Bayes.
3. **Find the base rate.** If nobody stated one, that is your missing input.
4. **Build a table of 10,000 cases**: prevalence splits the column, sensitivity and specificity split each row.
5. **Read the answer off the table** as a ratio of counts, not a formula.
6. **Check any multiplication** you did for a shared cause before trusting independence.
7. **If it is a quantity rather than an event**, report a mean *and* an SD, and simulate rather than plugging in averages.

How to attack any probability question, in order

1. **Name the events** precisely, in words, before any arithmetic.
2. **Write down which conditional you were given** and which one you were asked for. If they differ, you need Bayes.
3. **Find the base rate.** If nobody has mentioned one, that is your missing ingredient.
4. **Build a table of 10,000 cases** instead of manipulating formulas.
5. **Check the independence** of anything you multiplied. What would make these events move together?
6. **For a quantity, report a mean and a spread**, and simulate rather than plugging in averages.

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
Simulate any event and estimate its probability	`rng.random`, `rng.integers`
Exact binomial probabilities and tails	`stats.binom.pmf`, `.cdf`, `.sf`
Counts over a window (Poisson)	`stats.poisson.pmf`, `.sf`
Waiting times	`stats.expon`
A natural-frequency screening table	a few lines of arithmetic, no library
Mean and SD of a simulated quantity	`.mean()`, `.std(ddof=1)`
Expected cost or profit under uncertainty	simulate the scenarios, then average

If you cannot say what the output means, the fact that you produced it is worth nothing.

The traps, all in one place

Trap: the ones that bite most often

- Flipping a conditional: reporting $P(B \mid A)$ when the question asked $P(A \mid B)$.
- Ignoring the base rate after a positive test or a model flag.
- Multiplying probabilities of events that share a common cause.
- Believing chance “evens out” in the short run.
- Ranking small groups by rates and believing the extremes.
- Being amazed by a coincidence without counting the opportunities.

Ten questions to be able to answer

1. A 99%-accurate test for a 1-in-1000 disease comes back positive. Now what?
2. Explain the difference between $P(A \mid B)$ and $P(B \mid A)$ with an example.
3. What does independence require, and when have you seen it assumed wrongly?
4. A coin lands heads six times in a row. What is the chance of heads next?
5. Why is a 1% daily failure risk a serious problem over a quarter?
6. Define sensitivity, specificity, and predictive value, and say which one a patient wants.
7. Our fraud model flags 2% of transactions. How would you tell whether it is any good?
8. Why are the healthiest and the sickest counties both mostly small ones?
9. Three redundant servers each fail 1% of the time. What is your real risk?
10. Someone shows you a striking pattern in last month's data. What do you ask first?

Mastery checklist

You have this unit when you can, from memory:

- use complement, addition, and multiplication rules without confusing them
- say precisely what independence requires, and spot a shared cause
- translate any accuracy claim into a conditional probability, in the right direction
- solve a Bayes problem by counting 10,000 people
- name sensitivity, specificity, prevalence, and predictive value, and say which changes with the population
- explain why streaks, clusters, and coincidences are what randomness looks like

Where we go next

- Probability describes whether something happens. **Unit 8** attaches a *number* to the outcome, which is what lets us talk about an average, a total, and a risk.
- That is also the last piece we need before **Unit 9** explains why every standard error you have used since Unit 1 was legitimate.

Principle: the habit to keep

Before you explain a pattern, work out what pure chance would have produced.